

Simple Quiz Game for Kindergarten Students

Final Project for completing Basic Programming with Python Course in EDGE Program at Department of Information and Communication Technology, Mawlana Bhashani Science and Technology University.

Submitted by

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Description of the Project

This engaging and interactive quiz game is designed specifically for young learners, particularly students and children currently enrolled in kindergarten or equivalent educational institutions. The game aims to provide an enjoyable yet educational experience, helping children develop their cognitive skills while having fun. It is structured in a way that keeps the players entertained and motivated to participate actively.

The quiz consists of a total of 30 carefully curated questions, ensuring a diverse range of topics that are suitable for young minds. However, each game session will present only 20 randomly selected questions, providing a fresh and unique experience every time a student plays. This randomness ensures that no two gaming experiences are identical, maintaining the excitement and curiosity of the players. Additionally, to prevent repetition and maintain engagement, no question will appear more than once during a single playthrough.

To enhance user experience and clarity, the quiz displays the question number before presenting each new question. This feature helps players keep track of their progress by showing how many questions they have already answered and how many remain. Such a structure ensures that children can gauge their advancement, thereby adding a sense of accomplishment as they proceed through the game.

One of the standout features of this quiz game is its real-time feedback mechanism. If a player selects the wrong answer, the correct answer will be immediately displayed, reinforcing learning through instant correction. On the other hand, for every correct answer, a positive reinforcement message will appear, encouraging students and boosting their confidence. This dynamic response system makes the quiz highly educational while maintaining an enjoyable and engaging atmosphere.

Another user-friendly aspect of this game is the ability to exit at any stage. If a player wishes to quit, they can simply type "quit" instead of an answer, and the game will immediately terminate. At the end of the game—whether the player has completed all 20 questions or exited early—the final results will be displayed. The results will include the total number of correct answers and the corresponding score.

Scoring in this quiz follows a simple and structured format. Each correct answer carries a value of 10 marks, meaning that a player answering all 20 questions correctly can achieve a perfect score of 200. If a player decides to exit before completing all 20 questions, their score will be calculated based on the number of questions they have answered correctly out of the attempted ones, keeping the total possible score at 200. This approach provides a flexible and fair assessment system for all players.

To pass the quiz, a player must achieve at least 40% of the total score, which is equivalent to 80 marks. If a player meets or exceeds this threshold, they are considered to have successfully passed the test. However, if their score falls below 40%, a message displaying "Try Again" will appear, encouraging them to retry and improve their performance. This feature instills a sense of challenge and motivation, prompting children to strive for better results.

This quiz game has been meticulously developed to meet all the requirements outlined in the final project for Basic Programming with Python Course in EDGE Program at Department of Information and Communication Technology, Mawlana Bhashani Science and Technology University. It has been implemented using the Python programming language, ensuring simplicity and efficiency. The only external library utilized in this project is "random," specifically employing the "choice" function to select questions randomly.

Overall, this quiz game serves as an excellent interactive learning tool for kindergarten students. It combines education with entertainment, reinforcing knowledge retention through engaging gameplay. With its intuitive design, real-time feedback, and flexible scoring system, it encourages young learners to participate actively while gaining valuable knowledge in a fun and rewarding manner. This project successfully blends technology with early education, making it a valuable addition to the learning process of young children.